THE QUERYGRAPH STACK

Secure corporate agentic AI,
from governed prompt to audited answer

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querygraph.ai • firstpair.org



THE THESIS

Agents need a data plane that can say “no.”

MEANING

A semantic graph instead of raw tables

AUTHORITY

Capabilities travel with every action

EVIDENCE

Identity, lineage, and receipts survive

FOUR LAYERS

One governed semantic lakehouse

QUERYGRAPH	Agent-facing integration • memory • standards • evidence
TYPESEC	Identity • policy • capabilities • TypeDID • receipts
GRUST	Backend-neutral property graph • Cypher/GQL • semantic layer
LAKECAT + SAIL	Iceberg catalog • governed planning • Arrow/DataFusion execution

DEPENDENCIES POINT DOWN • EVIDENCE FLOWS UP

END TO END

A corporate question, not a database query

1. **Specialist agents**
   sales • legal • finance • HR
2. **Shared governed memory**
   facts persist across restart; authority does not leak

SUPERVISOR AGENT

“Which strategic accounts are at risk this quarter, and who may act?”

Must combine CRM, contracts, org structure, policy, and provenance.

PYDANTIC AI V2

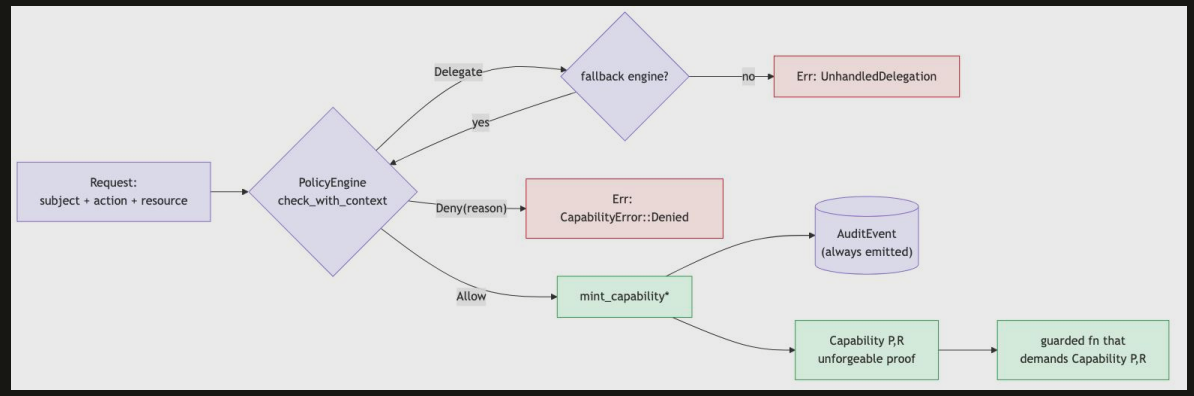
Ergonomics above. Capabilities below.



Private signing material stays in runtime deps — never in model instructions or tool args.

TYPESEC

Authorization becomes something code must carry



1 Policy decision

RBAC • ODRL • graph policy
WorkOS • Arcade • JWT

→

2 Capability<P, R>

subject + action + resource
explicit, attenuable authority

→

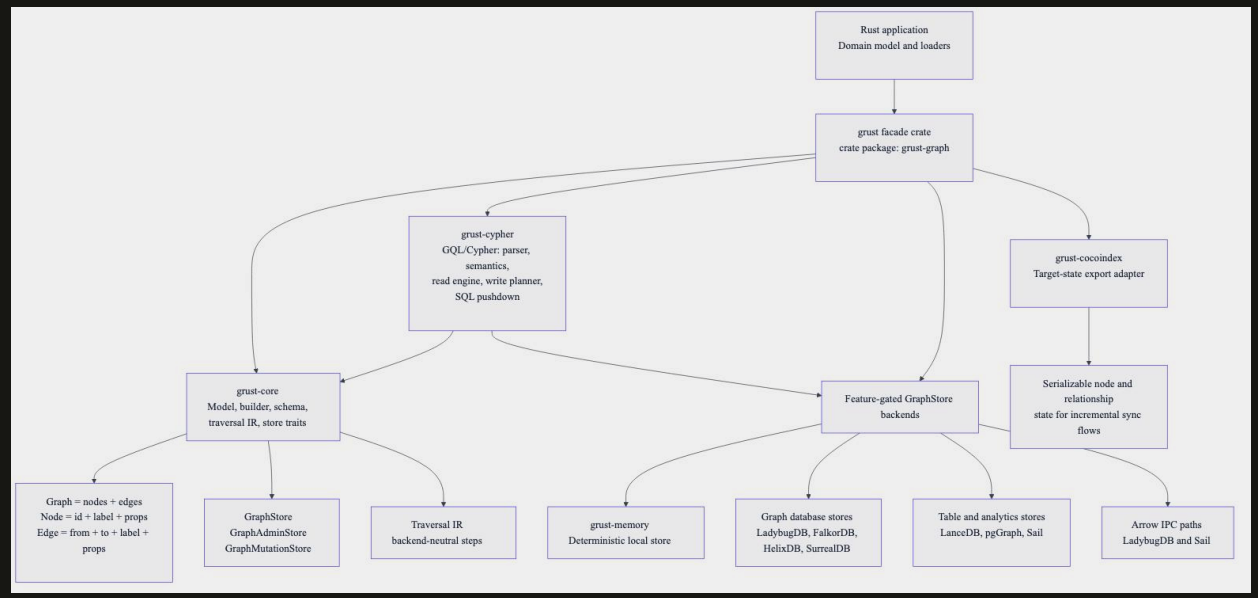
3 Protected tool

cannot execute unless
the capability is present

DENY BY DEFAULT • RECEIPT ON ALLOW • DIALECT-NATIVE DENIAL ON BLOCK

GRUST

The semantic layer is a graph, not a prompt



ONE MODEL
nodes • edges • typed properties

PORTABLE
memory • Turso • Sail • Postgres • graph stores

QUERYABLE
traversal IR • GQL/Cypher • Arrow interchange

Employee → reports_to → Manager

Account → governed_by → Contract

LAKECAT + SAIL

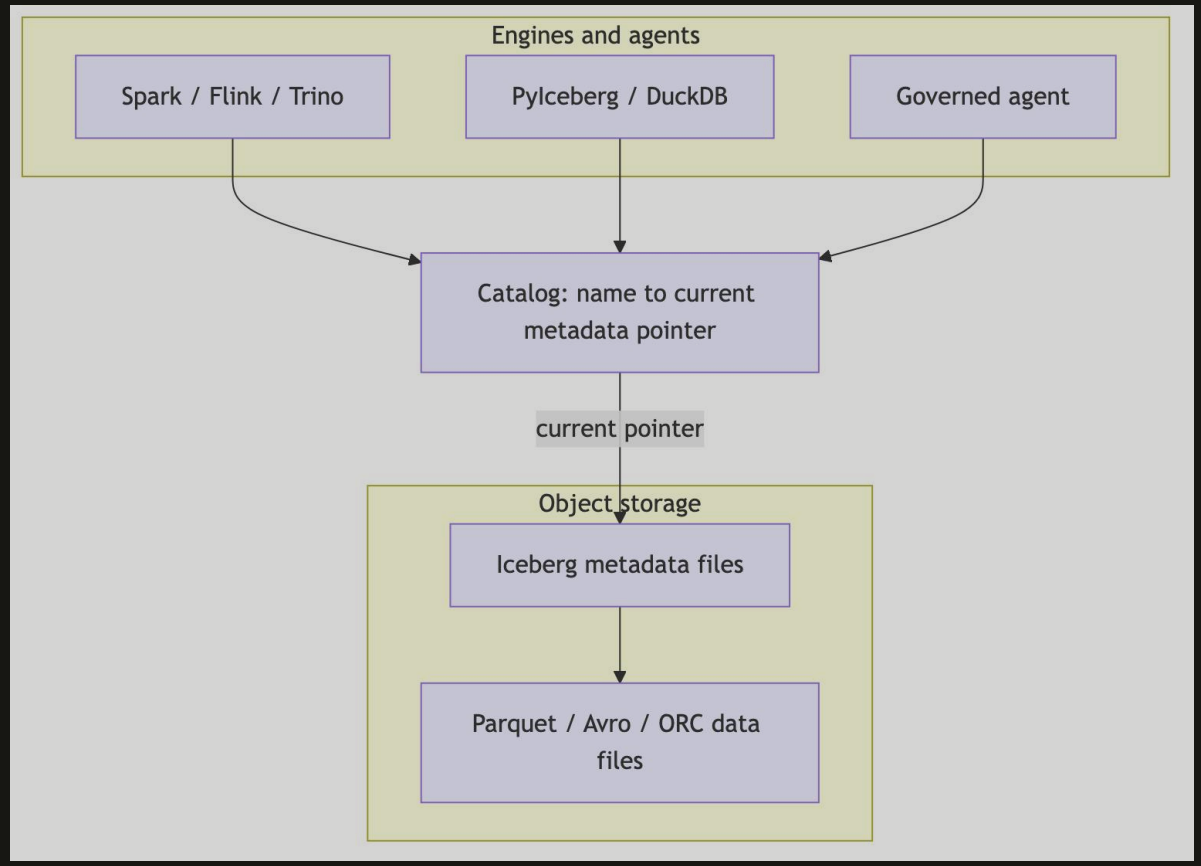
Keep the catalog thin. Bring planning to the data.

1 **LakeCat — durable control spine**

Iceberg REST • identity • pointer CAS • outbox • receipts

2 **Sail — JVM-free execution spine**

Spark Connect • Arrow • DataFusion • pruning • scans



Standard clients remain standard. Governed agents get a safer path.

THE CRITICAL FORK

Raw credentials are the exception

AUDITED EXCEPTION

RAW CREDENTIAL VENDING

Trusted human or explicit break-glass.
Scope and receipt are durable.

AGENT DEFAULT

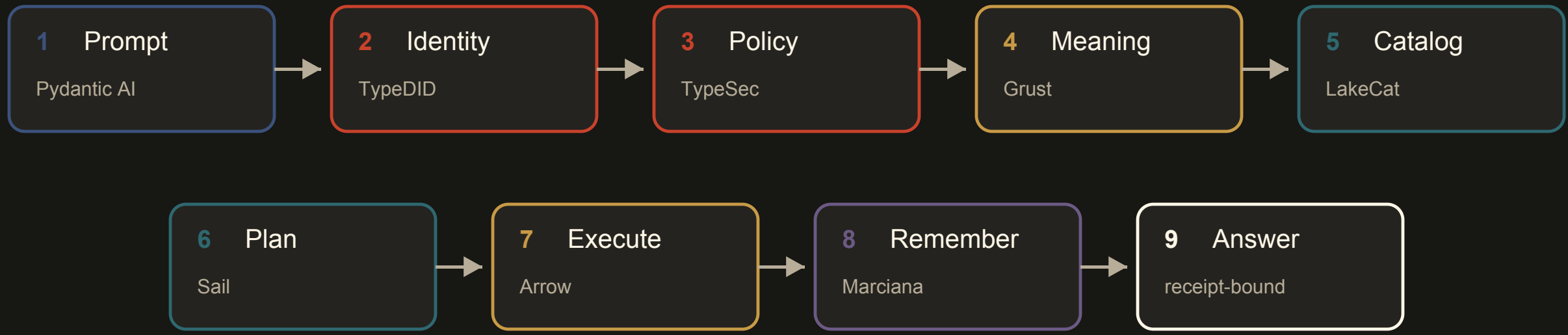
SAIL-PLANNED READ

Server-owned projection + row filters.
The policy becomes part of the plan.

The safest credential is the credential the model never receives.

ONE QUESTION

Nine typed boundaries



Every hop can be denied, attributed, replay-checked, and explained.

TYPEDID

Identity + intent + payload, bound in one envelope

SENDER / RECIPIENT
portable DID identities

CONVERSATION
task • session • room • mode

PROFILE
negotiated crypto suite

CIPHERTEXT
payload protected in transit

SIGNATURE
route • action • body digest

REPLY_TO
answer cannot be rebound

Transport stays transport: A2A • ACP • BAND • HTTPS • MCP

OLLAMA

Encrypted prompts. Capability-gated reveal.



Decryption is not authorization. TypeSec controls the reveal.

MARCIANA

Remembering is a commodity. Authority is not.

1 SPECIALIST WRITE
sales agent records a finding with provenance + purpose

2 PROCESS RESTART
memory survives; runtime credentials do not leak

3 SUPERVISOR RECALL
clearance + policy choose full, redacted, or denied



401
unsigned caller

403
signed outsider

THE ANSWER

Useful to the executive. Defensible to the auditor.

RISK BRIEF

3 accounts need intervention.
2 actions are permitted now.
1 requires legal approval.

- √ identity verified
- √ policy receipt attached
- √ plan carries row + column restrictions
- √ sources and lineage preserved
- √ memory read is purpose-bound

The output is not merely generated. It is governed, attributable, and replayable.

WHY THIS STACK

No single layer can make a corporate agent trustworthy



Fast • secure • decentralized agentic AI

PINAX

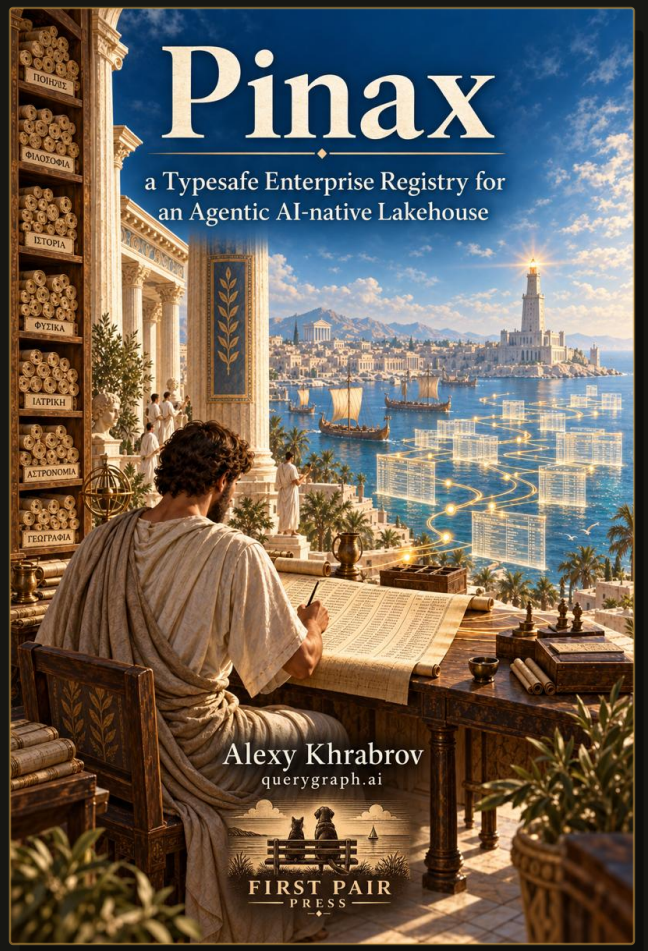
From business questions to shared meaning and data

Pinax is the agentic entrypoint to the QueryGraph stack: a typesafe registry above the catalog.

1 **TYPED TABLE CONTRACTS**
pinax.v1: profiles, typed columns, owners, classification, retention, allowed purposes

2 **GOVERNED DISCOVERY**
agents see only permitted tables and columns: snapshot-bound pages, no parallel RBAC

3 **SCHEMA-CHECKED PLANS**
TypeSec decides, LakeCat and Sail plan the scan; a plan is never an execution grant

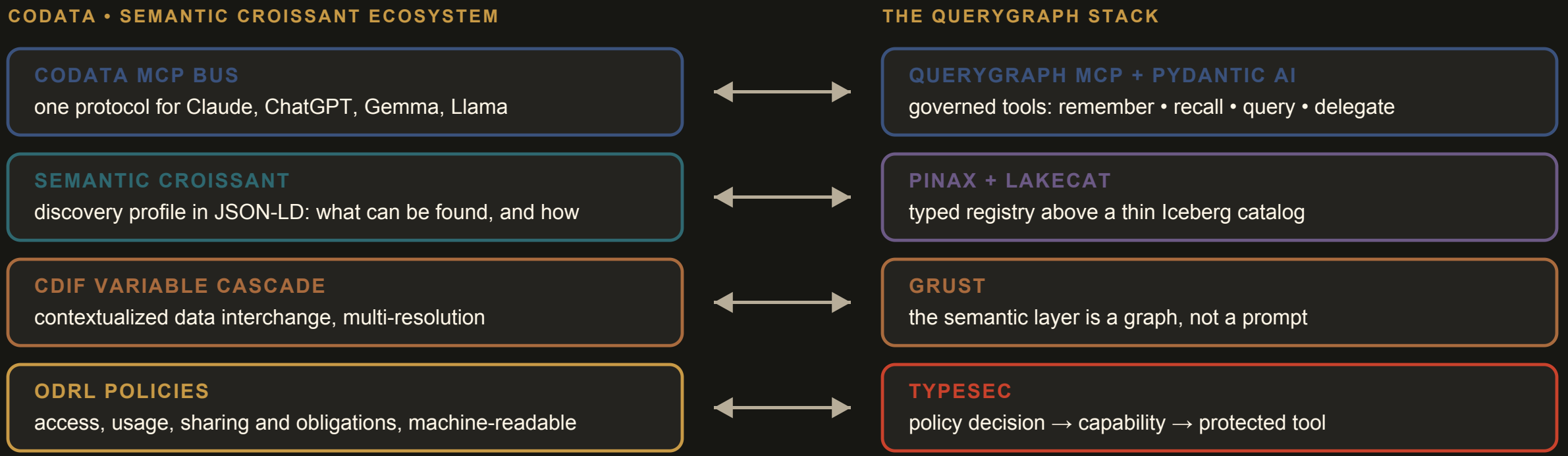


firstpair.org/books/pinax

PINAX OWNS MEANING • TYPESEC OWNS POLICY • LAKECAT OWNS THE CATALOG • SAIL OWNS THE PLAN

SEMANTIC CROISSANT

Beyond FAIR: research data that agents can act on



THE SAME FOUR PROBLEMS, SOLVED TWICE: FOR OPEN SCIENCE AND FOR THE ENTERPRISE

CDIF

CDIF: FAIR as data engineering, not bibliography

1. **A DEVELOPER'S GUIDE TO FAIR**
   domain-neutral standards, good web and data practice;
   the reference is the CDIF Book v1.1 at cdif.codata.org

2. **THE VARIABLE CASCADE**
   data and metadata at several resolutions, in JSON-LD;
   the research twin of Pinax typed table contracts

3. **CDIF4EOSC PLAYBOOK**
   profiles, guidelines, worked examples and code for EOSC;
   oceans • social science and climate adaptation • materials

FROM WORLDFAIR TO EOSC

first released May 2024 as a WorldFAIR output;
now the FAIR-integration playbook for EOSC

€8M	3 yrs	21
Horizon Europe	from June 2026	partners

EOSC nodes: DataTerra • Blue-Cloud • IOPAN

CDIF PROFILES DESCRIBE MEANING • PINAX CONTRACTS ENFORCE IT IN THE LAKEHOUSE

DID + ODRL

Identity and policy for research agents

DID: THE ADDRESS

decentralized identifiers with verifiable credentials

- persistent, resolvable, cryptographically verifiable, no central authority
- service endpoints say how to talk to the subject, human or agent
- the UNF-6 fingerprint is the seal: exactly the data the source published
- provenance and metadata stay with their owners

IN THE STACK: TYPEDID

identity + intent + payload bound in one signed envelope

ODRL: THE PERMISSION

open digital rights language policies

- policies for partners, users, agents, services and datasets
- ODRL client as an infrastructure component, driven from any IDE
- works for any model vendor; every agent gets its own identity
- ClimateAdapt4EOSC: legal and organizational interoperability

IN THE STACK: TYPESEC

policy decision → Capability<P, R> → protected tool

Every AI agent gets its own identity, and every action a receipt.

VAULT + MCP

From one-click deposit to a Time Machine

1 Collect
assistant reads any source, finds datasets and files

→

2 Describe
Croissant export for ML, CDIF variables enriched

→

3 Deposit
Dataverse via MCP, a DOI in one click

→

4 Review
human in the loop: Nectar, annotation, co-creation

→

5 Archive
Digital Vault with revisions, Signposting, Time Machine

WHERE THIS LANDS IN THE STACK

VAULT REVISIONS
LakeCat snapshot pointers, every version addressable

ASSISTANT MEMORY
Marciana: facts survive restart, authority does not leak

DEPOSIT LOG
TypeSec receipts: who did it, with which capability

Who actually did the work? The receipt says. mcp.dataverse.org • github.com/gdcc/mcp-dataverse

JOINT AGENDA

Open questions we answer together

Q Should AI agents have a digital identity?

Yes. TypeDID and DID give each agent a portable, verifiable identity.

Q Who is responsible when an agent does something wrong?

The receipt binds actor, capability and action; TypeSec denies by default.

Q Who owns data created by AI: user, organization, vendor?

Stewardship is declared in Pinax; rights and obligations in ODRL.

Q How accurate is AI-generated data?

Provenance in CDIF, UNF fingerprints, and a human in the loop before publish.

Q How do institutions, countries and continents interoperate?

Shared profiles and ODRL policies, federated catalogs, no central authority.

Q How do we shut access down and remove exposed data?

Capabilities attenuate and revoke; snapshot-bound plans; the Time Machine.

FAST • SECURE • DECENTRALIZED AGENTIC AI, FOR OPEN SCIENCE AND THE ENTERPRISE

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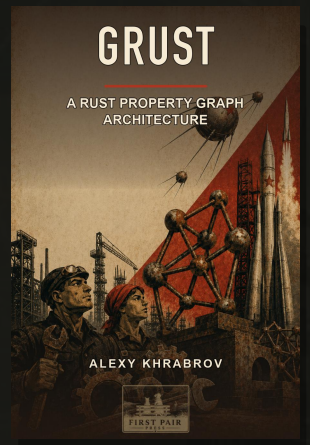
Books as living, source-linked systems.



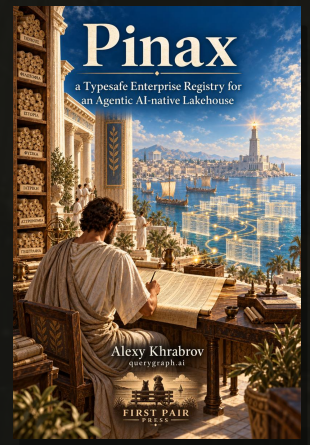
STACK



SECURITY



GRAPH



REGISTRY



CATALOG



ENGINE



STREAM

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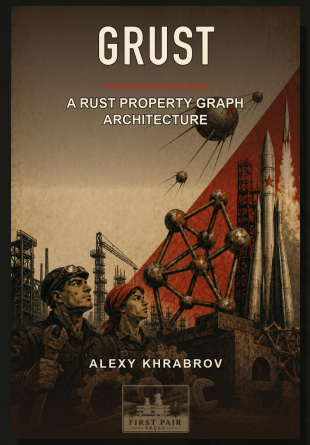
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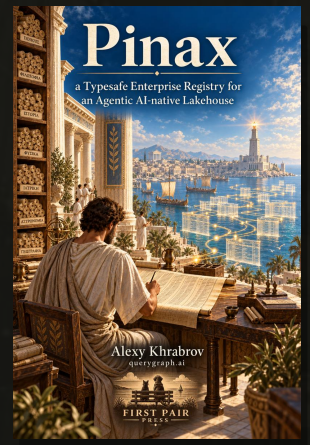
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